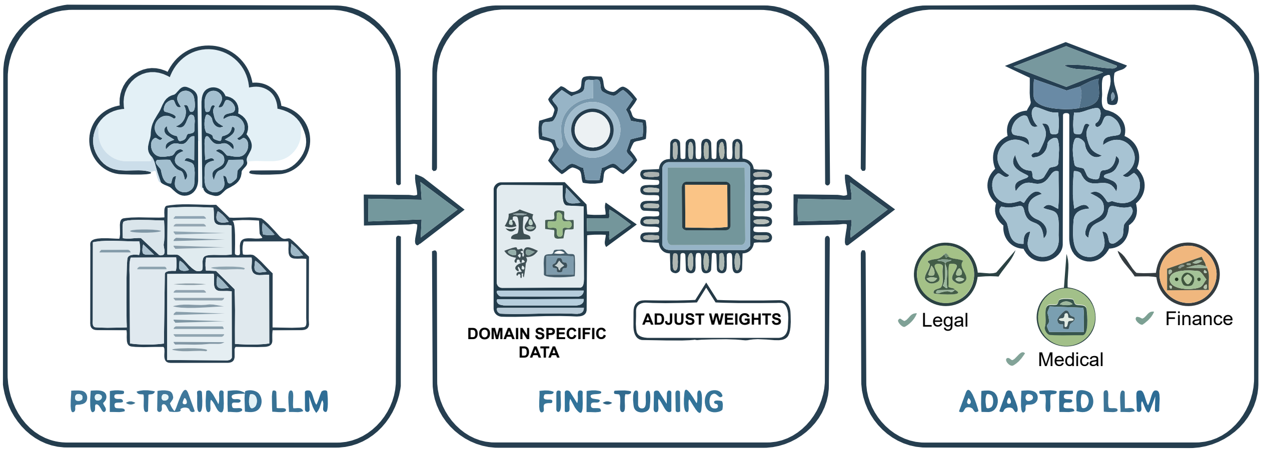
How to Fine-Tune LLMs for Domain-Specific Adaptation

In this first part of a two-part series, discover why we need to fine-tune large language models, and get introduced to the core concepts and techniques used for their domain-specific adaptation.



Large language models (LLMs) are deep neural networks, typically based on the Transformer architecture, trained on massive text corpora to predict the next token in a sequence. These models can generate text, write code, summarise documents, and answer complex questions. However, a general-purpose LLM cannot automatically perform well in specialised domains such as finance, healthcare, legal compliance, or enterprise-specific tasks. This limitation creates the need for LLM adaptation, where a pre-trained model is modified to behave correctly for domain-specific use cases.

Approaches to LLM adaptation

LLMs often need adaptation for two main reasons. The first is the limitations of base models -- pre-trained models possess generic knowledge and may produce inaccurate or contextually irrelevant responses (i.e., prone to hallucinations). The second reason is the need for customisation, as enterprises require outputs that follow internal standards — specific tone, formats, templates, and regulatory language.

There are three main approaches to adapting an LLM: prompt engineering, RAG, and fine-tuning.

Prompt engineering: Prompt engineering adjusts the model's behaviour through carefully crafted instructions without changing the model itself. It is best suited for quick behavioural adjustments in situations where the base model already has sufficient domain knowledge.

RAG (retrieval-augmented generation): RAG enriches the model's responses by retrieving relevant external documents and injecting them into the prompt at inference time. This enables the model to generate knowledge-based answers using fresh or domain-specific information without requiring retraining.

***Fine-tuning**:* Fine-tuning updates the model's parameters using domain-specific datasets, so it learns terminology, style, and task patterns internally. It is used for deep domain expertise or specialised tasks where consistency and precision are essential.

What is fine-tuning?

Fine-tuning involves taking a pre-trained LLM and further training it on a focused, domain-specific dataset. During fine-tuning, some or all the